

A Lifecourse and Individual-based View on Lifestyle to Enhance health (ALIVE)

1. General information

1.a Applicants

In case an applicant has more than one affiliation, please mention this in the table.

Name*	Institution	Department	Email	Tenured (i.e. fixed) position/ Tenure-track/ Other (please specify)	For Erasmus MC applicants: clinician/ researcher/ mix	Role in Flagship [Main Lead/Lead/Applicant]
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Corresponding Lead: Annemarie Luik

Planned start date of seed funding: 1-9-2022

Primary Convergence theme: *Improving Health Journeys*

Secondary Convergence theme (if applicable): *Human Centered Technology and AI for Health*

Tertiary Convergence theme (if applicable): *Technology Supported Transitions in Health(care)*

Quaternary Convergence theme (if applicable): Choose an item.

Keywords: lifestyle, health, prevention, lifecourse, individual-based approach, research infrastructure

1.b Summary (max 300 words)

Please briefly describe the objectives, design, and anticipated scientific, societal and economic impact for an academic audience.

Lifestyle factors such as physical activity, diet and sleep are crucial for an individual's health throughout life. To **ensure lifestyle interventions are more successful** at improving health, an individual-based approach should be adopted, taking into account an individual's diverse and changing characteristics, such as life stage and socio-economic status, and their attributes such as personal goals and preferences.

The **overall objective** of **ALIVE** is to contribute to the development of more effective, individual-based lifestyle interventions by establishing a research infrastructure, by focusing on individual characteristics and attributes across the lifecourse, and by developing strategies to improve existing interventions in order to maintain and/or acquire health. As combined knowledge, technologies and resources across academic disciplines are needed to achieve this objective, **ALIVE** is a collaboration of TU Delft, Erasmus MC and Erasmus University. Although our long-term aim is to provide the tools to improve any type of health, we will focus on brain and mental health within the first 5 years, creating transferable knowledge, skills and a research infrastructure.

Our **aims** are to:

1. Create a data science network to collect, combine and provide data to support cross-study (e.g., cohorts, registries, national databases) and multidisciplinary research collaborations.
2. Develop new and improve existing quantification methods for lifestyle behaviors across individuals using state-of-the-art device-employed measures to more precisely estimate associations with health.
3. Unravel the multidimensional interactions between lifestyle and their effects on maintaining and acquiring health across individuals over the lifecourse.
4. Design tools to identify individual attributes, such as personal goals and preferences regarding optimal health and lifestyle, as well as tools to address those attributes in health interventions.
5. (Re-)evaluate existing interventions and policies using a novel individual-based approach and new (combinations of) data to develop recommendations, guidelines for practitioners, and applications across health care and policy.

1.c Public Summary (max 300 words)

Layman's summary.

We weten allemaal dat leefstijlfactoren zoals fysieke activiteit, goede voeding en slaap cruciaal zijn voor een gezond leven. Toch is onze leefstijl suboptimaal en werken veel interventies om onze leefstijl te verbeteren niet goed. Leefstijlinterventies zijn alleen effectief als zij aansluiten bij de diverse karakteristieken, doelen, mogelijkheden en omstandigheden die mensen in verschillende fasen van hun leven hebben.

In **ALIVE** brengen wij experts op het gebied van gezondheid en technologie bij elkaar om bij te dragen aan effectievere leefstijlinterventies die op de specifieke persoon gericht zijn. Hiervoor creëren we een infrastructuur voor onderzoekers van de TU Delft, Erasmus MC en Erasmus Universiteit die het mogelijk maakt om individuele kenmerken en overtuigingen in de relatie tussen leefstijl en gezondheid gedurende de levensloop te verduidelijken en zo strategieën te

ontwikkelen waarmee bestaande interventies verbeterd kunnen worden. Op die manier dragen we bij aan een verbetering van leefstijlkeuzes en uiteindelijk een betere volksgezondheid.

Onze doelen zijn:

1. Het opzetten van een *data science* netwerk om reeds verzamelde gegevens uit cohort- en interventiestudies, registers en nationale databases te combineren, zodat we optimaal gebruik kunnen maken van bestaande data.
2. Het ontwikkelen van nieuwe en verbeteren van bestaande leefstijlindices met behulp van moderne technieken en apparaten, die inzetbaar zijn in verschillende individuen in verschillende fases van het leven.
3. Het onderzoeken van de multidimensionale complexe interacties tussen verschillende leefstijlfactoren en hun effecten op het behouden en bereiken van gezondheid gedurende het hele leven.
4. Het ontwikkelen van methodes om individuele overtuigingen (zoals persoonlijke doelen en voorkeuren) in kaart te brengen en mee te nemen in gezondheidsinterventies.
5. Het evalueren van bestaande interventies en beleidsmaatregelen, en het ontwikkelen van aanbevelingen, richtlijnen en applicaties die gebruikt kunnen worden door zorgprofessionals en beleidsmakers voor het verbeteren van leefstijl en gezondheid.

2. Relevance to Convergence Health & Technology mission

(50%) – max. 2500 words for 2.a to 2.e

- *The Flagship program contributes to the Convergence Health & Technology mission (1)*
- *The Flagship program objectives fit with the guidance per theme provided by the theme leads, and/or fit with more than one theme of Convergence Health & Technology (2)*
- *The Flagship program merges approaches and insights from distinct disciplines to reach its goals (3)*
- *The staff granted by this call actively brings together different disciplines within the program, and creates visibility and attracts talents, partners, and funding for the program (4)*
- *The Flagship program has a clear view on long term sustainability after the 5-year central funding period (5)*
- *The envisioned results of the Flagship program have the potential for a significant contribution to healthcare, society and/or have scientific impact which has the potential for long-term societal impact (6)*
- *The Flagship program provides opportunities for BSc/BA and/or MSc/MA students of the participating institutes to participate in (interdisciplinary) projects (7)*
- *The proposal defines the pathway of how this contribution can be realized. The pathway includes concrete research and/or impact pathway (societal and/or economical) valorization strategies (8)*

2.a Contribution to H&T mission and fit with H&T theme(s)

Although everyone knows that lifestyle is crucial for health, improving lifestyle has proven difficult. Current prevention measures and programs are suboptimal,^{1,2} because existing knowledge typically disregards individual characteristics, e.g., life stage and socio-economic status, and attributes, e.g., personal goals and preferences. To achieve an individual-based approach we need to understand what exactly constitutes **optimal lifestyle behaviors** and how these behaviors interact for different population subgroups and in different phases of life. This knowledge is currently limited by a lack of comparable data across population groups and by challenges in measurement and analysis methods to quantify these behaviors. Additionally, we need a better understanding of the **drivers of behavioral change** and effective ways to influence such change.

Impact on health

ALIVE will aid beneficial lifestyle behaviors to improve health. Doing so aligns with multiple goals of the Convergence Alliance. Ultimately, our ambition is to **improve individual health journeys** that ensure alignment with an individual's characteristics, e.g., their life stage and socio-economic status, and their attributes, e.g., personal goals and preferences. To this end we will form novel frameworks that foster scientific discovery and technological innovation, combining expertise and advances in the fields of technology, health and economics research to meaningfully target **lifestyle** as this is a crucial factor to **life-long health, from pre-birth to end of life**. We specifically emphasize the

importance of understanding and improving lifestyle factors from early life onwards and throughout life,³ as the dynamic and heterogeneous nature of lifestyle implies that the effects of lifestyle are different across stages of life and depend on individual characteristics and attributes.⁴ Moreover, as lifestyle affects us all, a better understanding of optimal lifestyle and its relation to health will allow **prevention to proactively maintain health**. Beyond **ALIVE's** focus on primary prevention, this program will also contribute to secondary and tertiary prevention. Additionally, by taking into account individual characteristics and attributes, we will contribute to **socio-economic equality in health** by determining which lifestyle works for whom at which life stage. Our program also links to **individual-tailored precise medical treatment for all** as our lifecourse- and individual-based approach to achieve 'precision' lifestyle interventions has relevant similarities in methods and approaches with individual precision-based approaches in medicine. To this end we will also collaborate with the NG-PM4all (Next Generation Personalised Medicine for all) Flagship.

Use of technology

Technology is vital to reach our ambition of better lifestyles and health for everyone. We will:

- Create a state-of-the-art data science network with integrated data from cohort studies, registries, and other datasets, which is easily accessible for researchers from various institutes and enhances reproducible science, collaboration and sharing knowledge.
- Combine expertise on different kinds and layers of data, from signal processing to environmental data, to ensure better use of existing data and to develop new quantification measures for the multidimensional factors of lifestyle.
- Develop and apply new (statistical) methods to create harmonizable data accounting for multiple sources and repeated measurements.
- Launch data-based (online) tools to support researchers, health care professionals and policy makers in designing, implementing and evaluating individual-based lifestyle interventions.

Together, this Flagship therefore also clearly fulfills the key aims of the H&T Health Journeys theme. We ensure **a more proactive and personalized whole-life approach to health and**

Definitions	
Lifestyle	• Human behaviors that are associated with health (RIVM). We focus on physical activity, diet and sleep.
Health	• The state of complete physical, mental and social well-being and not merely the absence of disease (WHO).
Wellbeing	• An individual's experience of their life as well as a comparison of the life circumstances with social norms and values (WHO), an integral part of the concept of health.
Brain health	• Encompasses neural development, plasticity, functioning, and recovery to optimize their cognitive, emotional, psychological and behavioral functioning (WHO).
Mental health	• A state of well-being in which an individual realizes his or her own abilities, can cope with the normal stresses of life, can work productively and is able to make a contribution to his or her community (WHO).
Individual-based approach	• An approach that takes into account the characteristics of individuals, e.g., their life stage and socio-economic status, and their attributes, e.g., their personal goals and preferences.
Research infrastructure	• The basic systems, equipment and software needed in order to work, operate and communicate effectively. This refers to the overall research infrastructure, including hardware, software, human capital, professional networks and education.

Figure 1. Definitions used in this proposal.

healthcare, by not merely assuming that the same prevention or treatment effort will work for everyone in each and every situation or life stage, but rather pro-actively use differences between people to be able to improve lifestyle and ensure **a longer time-span of healthy and active living**.

2.b Synergy between disciplines

Elaborate on how this program merges approaches and insights from distinct disciplines to reach its goals.

We will merge approaches and insights from **multiple disciplines and fields**: data science, technical developments in wearables and sensors, quantification methods, lifestyle and health sciences, economics, complex statistical analyses, behavioral change, and in developing recommendations, guidelines and applications/tools for practitioners. Therefore, we will work with a team of **highly skilled researchers** from different health disciplines (including researchers from Epidemiology and Biostatistics, General Practice, Rehabilitation Medicine and Public Health), economic and behavioral science (Erasmus School of Economics, Erasmus School of Health Policy & Management) and technology (Industrial Design Engineering, Sensor fusion and technology, Medical Informatics).

Together we have access to **multiple source data** from the Rotterdam area's most known population cohorts, such as the Rotterdam Study, the Generation R Study and Rijnmond Primary Care Database (Rijnmond Gezond); cohorts focusing on specific population groups, such as the Healthy Ageing and Intellectual Disabilities study (GOUD); and many clinical studies such as the Erasmus Stroke Study and the Co-Flow study. We also have access to various national (linked) registries, such as mortality registries and other data available through Statistics Netherlands (CBS).

2.c Long term sustainability plan

Please indicate your view on the feasibility of the program as well as the long-term sustainability plan of the program after the 5-year central funding period. Describe possible risks and how to tackle them.

Similar to the mission of the Convergence Alliance, we like to think big and improve any type of health. To ensure best use of resources, we will first focus on lifestyle and prolonging brain and mental health. This focus will allow us to create transferable knowledge, skills and a research infrastructure within the first 5 years, that can be extended to **any other type of health** afterwards. To ensure we are able to extend this approach long-term to any type of health, we focus on transferable skills and a widely applicable solution.

First, we will fully develop three main technological advances that can be directly applied and used by technologists, scientists and clinicians within TU Delft, Erasmus MC and Erasmus University, for various populations and health outcomes, and without further need of development, providing long-term sustainability after the initial 5-year period:

1. **Collaboratory of large databases**: This network will substantially lower the barriers for a joint analysis of individual-level participant data for researchers interested in lifestyle and health before the end of the first 5-year period (2027). These individual-level participant data will include the main cohorts of Erasmus MC and beyond (e.g., Rotterdam Study, Generation R, Rijnmond Gezond, GOUD),⁵⁻⁸ and possibilities to link with other data sources (e.g., socioeconomic data from CBS).
2. **Methods for data science**: We will publish open-access codes describing at least one improved and one new quantification method for assessing lifestyle (e.g., physical activity, diet, sleep) developed within **ALIVE**. Openly accessible codes to analyze complex data

from wearables and sensors are needed so that these can be applied in research and clinical settings throughout the world and future intervention programs to improve population's lifestyle and health. Additionally, we will publish open-access code describing at least one improved/new statistical method to create harmonizable data accounting for multiple sources and repeated measurements. These methods are needed to overcome biases related to different measurements and study designs. Additionally, we will publish at least one open-access code describing improved/new statistical methods to establish the complex interactions between multidimensional lifestyle factors, needed to study lifestyle as a whole. Publication of these codes will continue throughout the project duration.

Second, to enhance sustainability after the initial funding period, we have created a large team, spanning all partner institutions and career stages to facilitate convergence and make our Flagship futureproof. By involving a large number of talented assistant and associate professors, representing the future of the three institutions, we ensure the continuity and long-term sustainability of **ALIVE**. Moreover, our team is dedicated to teaching and mentoring so that the Convergence will also reach Bachelor and Master students, and at least 15 research internships will be established for interested students to inspire future generations.

Third, we will develop a general framework/guidelines for researchers to create effective interventions, rather than focus on creating a new intervention. Although many new lifestyle interventions are being developed, many run the risk of never being used in clinical practice due to a lack of technological advances or lack rigorous testing. Rather than adding to the pile of existing but unused or of insufficient-quality interventions we will develop a framework helping academia and industry to collaborate in adapting current or designing future effective interventions. We believe this will be more sustainable and have a bigger impact on both individuals and society.

Lastly, towards acquiring further funding in highly competitive national and international funding schemes, our team has an excellent track record, for example by being part of the Smarter Choices for Better Health Erasmus Initiative, and receiving funding from amongst others ZonMw, NWO, ERC, NIH and Alzheimer's Association for research questions of relevance to **ALIVE**, of which several still run over the upcoming years. Moreover, they are in close contact with industry partners (e.g., Philips, TNO) and health insurance companies (e.g., Menzis, Achmea) to ensure their support.

2.d Visibility, attracting talents and new partners

Describe how you will create visibility for the flagship program, how you will engage students (BSc/BA and/ or MSc/MA) and early career researchers in interdisciplinary projects, as well as how you plan to attract talents, partners and funding.

Our consortium members are involved in **education** at the BSc, MSc and PhD levels at the three institutions, e.g., Medicine, Health Sciences, Health Economics, Economics, Rehabilitation Medicine, General Practice, Intellectual Disability Medicine, Industrial Design Engineering, and Systems&Control. They are also already involved in education across institutes, e.g., Erasmus MC researchers in the Clinical Technology program at TU Delft or Erasmus University researchers in the Health Sciences program of Erasmus MC. This will enhance visibility of **ALIVE** and offer the opportunity to engage talented and diverse young researchers in our program. We will offer students the unique opportunity to develop themselves in the fields of both technology and health sciences and collaborate with fellow students from other institutes. As these fields are expected to integrate more and more in the future, this will be an interesting and promising area

to work in. Beyond courses, in the first 5-year period of **ALIVE** we will also offer at least 15 MSc **research internships or thesis projects** for students from TU Delft, Erasmus MC and Erasmus University.

Our consortium members' visibility is also evident from the many presentations of our work at large national and international conferences, talks for TEDx and the 'Universiteit van Nederland'. Also, our work reaches the wider public via a wide range of regional and national (media) channels such as Radio Rijnmond, AD, Trouw, NRC and NOS. We are actively involved as board and committee members in local, national and international research networks related to lifestyle and health, such as the Physical Activity Community Erasmus MC (PACE), the European Sleep Research Society, the Dutch Academy of Nutrition Sciences, Health Systems Design Research Group, International Society For Physical Activity And Health (ISPAH), and the Alzheimer's Association, ensuring future visibility of the **ALIVE** Flagship. We will exploit our involvement in these networks to initiate new projects with partners around the world.

Next, we create impact and visibility by being involved in translating research findings to applications, e.g., in developing guidelines on nutrition and prevention of eating disorders within the Dutch health council, and developing and communicating physical activity and lifestyle guidelines for cardiac patients and for people with intellectual disabilities. Also, we have been involved in the successful development of digital tools and applications, such as Sleepio (www.bighealth.com), a digital treatment for insomnia used by multiple international companies and the NHS Scotland for example, and Get Wise about Your Health, an online platform that provides education on lifestyle and health for people with intellectual disabilities.

As a team we have a broad network of other universities, health foundations, health insurers and policy makers (e.g., municipality of Rotterdam, Dutch Health Council, Dutch Healthcare Authority (NZA), health insurance companies, industry partners). We will use these external partners as a sounding board, to secure future funding and to make sure our research findings reach and are compatible with stakeholders and individuals in society.

2.e Long term societal impact

Describe the potential and possible pathways for a significant contribution to healthcare, society and/or scientific impact which has the potential for long-term societal impact. Pathways include concrete research and/or impact pathway (societal and/or economical) valorization strategies. If applicable, indicate expected TRLs and necessary partners. In case of fundamental research please describe the scientific impact and time scale of expected societal impact.

As lifestyle is key in maintaining and acquiring health across the lifecourse, improving lifestyle will have major impact on multiple levels. Here we summarize the most important examples of expected scientific and societal impact of **ALIVE**:

Scientific impact:

- A data science network to collate information from a range of studies and registries, forming an interface that can be used by any researcher of the participating institutions to analyze health data from large populations. This will boost convergence, further (inter)national collaborations, but also open and reproducible science. The data science network will allow to generate findings about health and health care that cannot be generated with any of the individual datasets alone as we:
 - link novel combinations of data, e.g., socioeconomic, genomic and imaging data of individuals with different characteristics and in different life stages.

- have the expertise and ability to account for confounding and differences in the (quality of the) study designs.
- have greater statistical power.
- A continuous interaction between health, policy and technology researchers of different institutes to ensure long-term convergence and cutting across disciplines, leading to:
 - publication of findings in high-impact scientific open access journals with authors from all institutions.
 - yearly Flagship program symposia open to all researchers.
- Improved and new quantification methods that can be used by other research institutions, therewith progressing the broader scientific field.
- A framework that pinpoints which elements of interventions fit best with specific individuals at specific phases in life, and what broader outcomes (e.g., education, participation) are relevant. The framework allows to re-evaluate and improve existing interventions and develop new effective interventions. Improved lifestyle interventions will have a sustainable impact on health throughout life.

Societal impact:

- Translation of findings to practical recommendations and guidelines that allow health professionals, policy makers and the general public to use advice tailored at the individual level, based on individuals' characteristics and attributes, and linked to societal outcomes most relevant to the individual during a certain stage of life.
- Development of a gold-standard framework and data-based (online) tools to support researchers, health care professionals and policy makers in designing and implementing effective individual-based interventions.
- A widely used and implemented framework to enhance use of the individual-based approach in clinical practice, interventions and policies by active cooperation with practitioners (e.g., GPs, physiotherapists), policy makers (e.g., health insurers, municipality) and other external partners.

Ultimately, our long-term impact will be better health for individuals throughout life, and reduced health care use and health care costs for society, by enhanced effectiveness of lifestyle interventions.

3. Flagship proposal (25%) max 2000 words for 3.a to 3.d

- The Flagship program contributes to excellent Convergence Health & Technology research, at the forefront of international research (1)
- The Flagship program is both innovative as well as challenging in its research and approach (2)
- The Flagship program has a clear strategy to engage students in its research plan (3)
- The Flagship program has a clear aim and long-term work plan, including, if applicable, a specification of required facilities and a data-and infrastructure plan. Coherence within the program is clearly demonstrated (4)
- The Flagship program aligns with or builds on (inter)national research agendas in this field, fit with the (inter)national research agendas but has a “high risk, high gain” main objective that is thoroughly explained (5)
- The Flagship program is feasible in terms of research performed, staff and budget needed (6)
- The Flagship program has a concrete plan for attracting external funding, which should result in at least the same amount as the central funding over the 5-year period (7)

3.a Ambitions

Describe how the ambitions of the flagship program are in line with Convergence Health & Tech research. Please elaborate on the innovative and challenging aspects of the research program and/or its approach.

Ultimately, our ambition is to **improve individual health journeys** that ensure alignments with individual characteristics, e.g., their life stage and socio-economic status, and their attributes, e.g., personal goals and preferences. More specifically, within the first 5 years of **ALIVE** we will contribute to the development of more effective, individual-based lifestyle interventions by establishing a research infrastructure, focusing on individual characteristics and attributes across the lifecourse, and by developing strategies to improve existing interventions in order to maintain and/or acquire brain and mental health. Key to our approach is the notion that for lifestyle interventions to be successful, they should be aimed at those aspects of lifestyle and health that are most important for individuals across different life stages.⁹ Individuals face different health risks at different stages of their lives,^{10, 11} and have different goals in terms of their functioning, participation and health.⁹ Furthermore, what lifestyle changes are most feasible or preferred, or have most impact on health also differs per individual.¹² We will use **innovative quantification measures, statistical methods and data sources** that will be combined and integrated in a **state-of-the-art data science network** to achieve individual-based knowledge of lifestyle and health. The data science network will be used to lower the barriers for researchers to better use, link and share datasets. It will enable them to answer questions about lifestyle and health that currently cannot be answered, or to answer them in new ways. **ALIVE** therefore provides new opportunities to **explore interactions between lifestyle behaviors and their effects on health** and the **development of individual-based tools** to enhance the effectiveness of lifestyle interventions. This knowledge and these tools will be translated into **practical recommendations and guidelines** for policy and interventions.

3.b Aims and feasibility

Describe the program's aims, including a clear strategy to involve students in its research plan. Elaborate on the program's feasibility in terms of research performed, staff and budget needed. Address a concrete plan for attracting external funding. Also address possible risks and how to tackle these.

The **overall objective** of **ALIVE** is to contribute to the development of more effective, individual-based lifestyle interventions by establishing a research infrastructure, by focusing on individual characteristics and attributes across the lifecourse, and by developing strategies to improve existing interventions in order to maintain and/or acquire health. Together this will enable the multidisciplinary study of lifestyle and health within TU Delft, Erasmus MC and Erasmus University. This project will provide knowledge, data and tools to facilitate the implementation of individual-based (lifestyle) intervention and prevention, focusing on those aspects of health and lifestyle that are important to different individuals themselves across different phases of life.

Towards this objective, we have 5 aims, yielding building blocks that can be used by researchers across the institutions:

1. Create a data science network to collect, combine and provide data to support cross-study (e.g., cohorts, registries, national databases) and multidisciplinary research collaborations. **(WP: Data Science)**
2. Develop new and improve existing quantification methods for lifestyle behaviors across individuals using state-of-the-art device-employed measures to more precisely estimate associations with health. **(WP: Quantification and Measures)**

3. Unravel the multidimensional interactions between lifestyle and their effects on maintaining and acquiring health across individuals over the lifecourse. (**WP: Health Associations**)
4. Design tools to identify individual attributes, such as personal goals and preferences regarding optimal health and lifestyle, as well as tools to address those attributes in health interventions. (**WP: Designing Change**)
5. (Re-)evaluate existing interventions and policies using a novel individual-based approach and new (combinations of) data to develop recommendations, guidelines for practitioners, and applications across health care and policy. (**WP: Evaluation and Recommendations**)

Feasibility

We ensure feasibility and tackle potential risks in multiple ways:

- We will make use of multiple data sources with large amounts of data available (population-based cohorts, clinical cohorts, GP data and registry data). Combining and integrating these data will allow us to perform studies from an overarching and multidimensional perspective, for example by looking at different types of lifestyle behaviors, complex measures of these lifestyle behaviors, and repeated measures across the lifecourse. As we focus on optimizing the use of already available data rather than new data collection, feasibility of the program is enhanced.
- We have external funding available for the first two years of the program (>2 million), and see potential for various grant proposals for the next years in the field of prevention, health and technology that fit research agendas as mentioned also in part 3.d.
- To mitigate potential risks related to making data available for different institutes, legal restrictions on combining data, and potential challenges due the computational complexity of algorithms, we will actively involve principal investigators (PIs) and data scientists among other experts. Additionally, we will involve the respective privacy and security offices.
- To prepare of rising price levels for e.g., resources and software, we have reserved additional budget.
- To mitigate loss of expertise if experts retire or move, our consortium consists of young leads and will include many students and early career researchers, emphasizing the long-term commitment to this project. Also, we believe that convergence between technology and health will become natural in the future and that these fields will merge, specifically when MSc and PhD students will be educated in combined expertise.

3.c Work plan

Elaborate on the long-term work plan, including, if applicable, a specification of required facilities and a data-and infrastructure plan. Pay attention to overall coherence of the work plan.

ALIVE consists of 5 highly connected work packages (WPs) that will be executed between the three institutes in close collaboration (Figure 2). To ensure true convergence, each WP includes members of at least two institutes and multiple departments. Below we summarize each of the WPs:

WP Data Science

The data science network is a main building block that will be exploited in all WPs, aiming to contribute to better use of existing infrastructures. Specifically, this WP will:

- Help Flagship researchers to identify, get access, combine and link the datasets needed for their projects.
- Help researchers to identify and if needed develop appropriate methodology to combine different data sources with varying populations (for example, different age groups) and study designs, with adjustment for any biases.
- Develop a business plan to make the knowledge that is acquired and centralized within this Flagship a viable service that may be provided to researchers of TU Delft, Erasmus MC and Erasmus University in the long-term.
- Actively seek collaborations with other Flagship programs with similar ambitions.

WP Quantification and Methods

Current methods used in cohort studies to quantify lifestyle and health are suboptimal in the sense that they only use a small portion of the available information that can be extracted from the data.¹³ This may contribute, for example, to false negative results (e.g., no relationships between lifestyle and health outcomes may have been found, while with advanced analyses these would be found in the whole population or for specific sub-groups). This WP will:

- Extract more information from the vast amount of accelerometer data, that goes beyond the currently applied quantification measures, such as fragmentation, quality and timing.
- Develop new data fusion algorithms in order to optimize use and interpretation of accelerometer data, e.g., when combined with context information such as geographic location or social environment. Therefore, in this WP new data algorithms will be developed to fuse data from different sources such as demographic data, clinical data and sensor data, e.g., based on smartphone data.
- Use these new quantifications of lifestyle to provide additional features, e.g., based on context information. These features will be used in data algorithms such as neural networks to improve prediction of the states of health.

WP Health Associations

Different lifestyle factors are interacting in their associations with health across the lifecourse.¹⁴ These lifestyle factors need to be addressed as multidimensional constructs, e.g., the construct of



Figure 2. Connectedness within the ALIVE Flagship.

physical activity in relation to health across the lifecourse consists of multiple dimensions: type of physical activity, frequency, duration and intensity.¹⁵ This WP will:

- Identify multidimensional lifestyle patterns by establishing the interactions of the different lifestyle behaviors (e.g., physical activity, sleep, diet) and their components (e.g., duration, fragmentation, quality, timing) across individuals over the lifecourse. Combined data from multiple data sources will be used to perform clustering and network analyses integrating multiple measures of different lifestyle behaviors over the lifecourse.^{16, 17}
- Investigate associations of these multidimensional lifestyle patterns with health outcomes across the lifecourse, specifically focusing on brain and mental health. Statistical methods that appropriately account for repeated measures and combined data sources will be used.

WP Designing Change

Health associations should be assessed from an individual-based perspective, taking into account an individual's characteristics, e.g., life stage and socio-economic status, and attributes, e.g., personal goals and preferences.⁴ How do individual-based attributes influence health and lifestyle, and how do individuals perceive the link between lifestyle and health? Could this perception explain (part of) the gap between intention and actual behavior? This WP will:

- Explore how individual characteristics and attributes influence lifestyle behaviors and health. A combination of human-centered design methods, experiments and data from existing cohorts, registrations and surveys will be used.
- Explore the role of health perception in determining lifestyle. Beliefs and expectations about the consequences of lifestyle might affect intention to change and therewith the effectiveness of lifestyle interventions. Similarly, some people might perceive effortful and persistent health behavior and health as competing goals, influencing their choices for a healthy lifestyle.
- Design tools to identify attributes, e.g., personal goals, preferences and beliefs of individuals, and tools to address these attributes in existing health interventions. This will be an important step towards designing personalized and effective interventions.

WP Evaluation and Recommendations

How do we to implement an individual-based approach to lifestyle in practice? To answer this question, the knowledge, data infrastructure and digital tools from all WPs will be synthesized and integrated. In close collaboration with partners from the field (municipalities, practitioners, health insurance companies), this WP will:

- Gain insight into and formulate advice on the best ways to take individuals' heterogeneous characteristics, e.g., stage of life and socio-economic status, and attributes, e.g., personal goals and preferences, into account in the content and design of health interventions and the way that interventions are offered to individuals (or individuals are 'led' towards interventions). This will help improve uptake and effects, specifically among vulnerable groups.
- Evaluate implementation strategies for diverse groups (e.g., stage of life, socio-economic status), using the infrastructure/database of the Flagship and develop strategies for gaps in reach and effects for diverse groups.
- Evaluate the effect of (existing or still running) lifestyle interventions and health policies across the lifecourse and across different groups on health, and societal outcomes using the data science network developed in [ALIVE](#).
- Adjust, develop and integrate new recommendations and guidelines for health care professionals on how to optimally design and lead people to interventions using those

aspects of health and lifestyle that are most important, in terms of individual characteristics, e.g., stage of life and socio-economic status, and attributes, e.g., personal goals and preferences.

- Develop and integrate materials for healthcare professionals, patients and public on the diverse relationship between lifestyle and health across individuals.

3.d Alignment of the program with relevant national and international research agendas

Explain how the program aligns with or builds on national and international research agendas in its fields or, if it does not fit with these agendas, elaborate on what is the “high risk, high gain” objective.

ALIVE emphasizes the importance of maintaining and acquiring health across the lifecourse through a healthy lifestyle. This is well-aligned with a general shift in focus from cure to prevention in society, which is also reflected in the agenda of many funding bodies. Similarly, overall health and wellbeing – compared to a focus on physical health or one disease only – is getting more and more traction and research agendas. Specifically, **ALIVE** aligns with the following national and international research agendas in the field:

- **KIA2020-2023** (knowledge and innovation agenda for creative industry) describes tools that will be applied to explore factors related to the intention-behavior gap.
- **NWO Transitions and Behavior**, focusing on societal challenges where change in behavior is required.
- **HORIZON-HLTH**, with e.g., current specific calls related to prevention of obesity across the lifecourse.
- **Manifest Hoofdzaken**, initiated by MIND, Hersenstichting and ZonMw, aiming to develop a national knowledge and innovation platform on brain and mental health.
- **Joint Programming Initiative a Healthy Diet for a Healthy Life (JPI HDHL)**, that funds collaborative new research to prevent or minimize diet-related chronic diseases.
- Letter to the parliament '**Nederland vitaal en in beweging**' in which the Ministry of Health, Welfare and Sport outlines its ambitions to promote exercise among the population.

4. Consortium (25%) max 2000 words for 4.a to 4.d

- The Flagship program is driven and executed by a well-organized team of scientists from all three institutions and at least 4 departments or faculties in total. In exceptional cases, it is possible to form a Flagship program with only two leads from two institutions, but this should be clearly motivated (1)
- Members of the consortium have a track record of excellence in academic research, translation of scientific results into clinical application or societal impact (2)
- Members of the consortium have a significant track record in engaging and supervising BSc and MSc students in research projects (3)
- Members of the consortium have demonstrated to be able to attract external funding, thereby enhancing the programs sustainability after the 5 years of central funding (4)
- There is a demonstrable fit between the aims and objectives of the program and the expertise of the consortium members (5)
- End users (private and/or public sectors) are involved from an early start and team members are well-connected to external partners in the ecosystem (6)
- Governance should be clearly described and is organized in such a way that it facilitates collaboration and optimally makes use of the expertise of all participants (7)

4.a Track record of the consortium members

Elaborate on the consortium members' track record in academic research, translation of scientific results into clinical application or societal impact, BSc and MSc supervision and engagement in research projects, ability to attract external funding after the 5-year central funding period.

ALIVE is led by four¹ excellent and ambitious researchers from the three institutes:

- **Annemarie Luik** is an assistant professor at the Departments of Epidemiology and of Child and Adolescent Psychiatry/Psychology at Erasmus MC. She has a double degree in epidemiology and psychology and received her PhD in 2015 for her work on sleep and mental health. She has worked at the University of Manchester and Oxford, before returning to Erasmus MC in 2018 to lead the Psychiatric Epidemiology research group. Her group's research focuses on sleep, mental health and neurological disorders across the lifecourse. She is member of the boards of the Dutch and European sleep societies. She has published over 80 papers in international peer-reviewed journals (key publications)¹⁸⁻²⁰).
- **Jos Kraal** is assistant professor at the Department of Human-Centered Design, faculty of Industrial Design Engineering at TU Delft. He holds a MSc in human movement sciences and a MSc in public health, and attained his PhD in medical informatics at the University of Amsterdam. After working on optimizing digital health interventions at the Máxima Medical Center in Eindhoven as a researcher and project leader, he joined TU Delft as an assistant professor as part of the 4TU Pride and Prejudice consortium. He is currently working on the intersect of behavioral science and human-centered design research to develop tools and strategies to improve the effectiveness of lifestyle interventions. Jos is part of the Health Systems Design Research committee, a special interest group of the Design Society working on system design for healthcare. He published over 20 papers in international peer-reviewed journals (key publications²¹⁻²³).
- **Trudy Voortman** is an associate professor at the Department of Epidemiology at Erasmus MC where she leads the Nutrition and Lifestyle Epidemiology group. She is principal investigator of nutrition and lifestyle in the Rotterdam Study and Generation R Study cohorts. Trudy has a background in nutrition sciences (MSc) and epidemiology (MSc and PhD) and previously worked as (guest) researcher at TNO, UC Berkeley, Unilever, and Harvard School of Public Health. She currently also leads the research integrity committee of the Dutch Academy of Nutritional Science, is part of the leadership team of the American Society of Nutrition, associate editor of Clinical Nutrition (top-10% journal in the field of nutrition) and is member of work groups of the Dutch Health Council. She has published over 100 papers in international peer-reviewed journals (key publications²⁴⁻²⁶).
- **Bram Wouterse** is an assistant professor in health economics at the Erasmus School of Health Policy & Management. His research focuses on socioeconomic inequalities in health and health care use, the relation between health and health care costs across the lifecourse, the fiscal consequences of health care expenditure growth, and long-term care. He is co-lead of the Health Equity line of the Smarter Choices for Better Health initiative and a research fellow at Netspar. Prior to joining Erasmus University, he was associated with the National Institute for Public Health and the Environment (RIVM), and the CPB Netherlands Bureau for Economic Policy Analysis. He has published 18

¹ An exception for four leads has been granted by the Convergence team for this proposal. Together these four early/mid-career leads have the expertise, knowledge and skillset at the main areas important to the **ALIVE** Flagship, essential to make the Flagship successful and sustainable.

international peer-reviewed papers in prominent (field) journals like PNAS, the Journal of Health Economics, and Health Economics (key publications ²⁷⁻²⁹).

The WPs will be led by the team leads together with **Teresa Bago d'Uva**,³⁰⁻³² **Pieter Bakx**^{27, 33, 34} (Erasmus University), **Daniel Bos**,³⁵⁻³⁷ **Hans Bussmann**,³⁸⁻⁴⁰ **Dieuwke Schiphof**⁴¹⁻⁴³ (Erasmus MC), and **Manon Kok**⁴⁴⁻⁴⁶ (TU Delft). In addition to the team and WP leads, the consortium consists of a wide variety of highly skilled, talented and ambitious researchers. Together, the **ALIVE** team therefore has an impressive track record as shown by:

- Key publications in scientific journals with high impact in their fields, such as *JAMA*, *NEJM*, *The Lancet Psychiatry*, *Psychiatry*, *Journal of Health Economics*, *JACC*, *PNAS*, *Statistical Methods in Medical Research* and many others, covering the topics of the **ALIVE** Flagship. Publications are well-cited and have been presented at conferences and university departments across the world.
- Translation of scientific results into clinic and society. For example, the development of national guidelines and policy evaluations (NZa, WHO, OECD) and research agendas (NWO), education for health professionals (books, seminars, e-learning), and digital applications and platforms (Sleepio, Artrose Gezond, Activity Monitoring, Slim Gezond, Get Wise about Your Health). In addition, our consortium members are dedicated to public engagement. They are participating in specific activities geared to learn children about science (Wetenschapsknooppunt, Junior Med School) and making sure their research reaches the lay audience via several formal media outlets (NOS, Algemeen Dagblad, NRC, Time Magazine), videos and podcasts (TEDx, Universiteit van Nederland) and social media.
- Attraction of internal and external funding from national and international funding sources as ZonMw, NWO, ERC, NIH and Innovative Medicine Initiative, which sets the stage for successfully attracting additional funding for the **ALIVE** Flagship. Multiple aspects of the current proposal were already covered by existing funding (>2 million), for example the multidimensionality of lifestyle behaviors, e.g., sleep health (NIH, Annemarie Luik), lifestyle intervention and implementation research (ZonMw, Marienke van Middelkoop; ZonMw, Dieuwke Schiphof), lifestyle in population subgroups, e.g., patients with chronic diseases or individuals with intellectual disabilities (ZonMw, Marienke van Middelkoop; ZonMw, Alyt Oppewal), lifecourse perspective research (Erasmus Trustfonds, Bram Wouterse and Pieter Bakx), data networks (Innovative Medicines Initiative, Peter Rijnbeek), sensor data and sensor fusion (ZonMw, Hans Bussmann; NWO, Manon Kok), and brain health and mental health (EU Joint Programme Neurodegenerative Diseases, Trudy Voortman; Alzheimers Association, Daniël Bos; Erasmus Trustfonds, Bram Wouterse).
- Excellent teaching, coordination and development of courses for BSc and MSc students on Behavioral Science, Industrial Design, Medicine, Health Sciences, Econometrics, and Computer Science across the three universities. In addition, they have extensive experience in the successful supervision of students at BSc, MSc and PhD level and served as (co)promotor to many now successful researchers in and outside of academia. Lastly, they also put student wellbeing at the forefront by coaching students and organizing wellbeing seminars and support programs.
- Representation in national and international scientific and societal committees. Our members are involved in health and societal committees such as the Dutch Health Council, NZa and European Medicine Agency. Additionally, they are involved in hosting international scientific conferences and workshops on a variety of topics such as Health System Design, Measurement of Physical Behavior, and Smarter Choices for Better Health.

Most importantly, this proposal will allow us to add new members to our team as the funding supplied by the Convergence Alliance will be mostly used to attract new talents to our institutes. We plan to hire 5 new PhD students to work across the WPs. Additionally, a combined position of post-doctoral researcher/data manager to support development of the data science network and statistical consultation, and a combined post-doctoral researcher/project manager to support overall coherence and convergence will be advertised.

4.b Expertise of the consortium members in relation to the aims and objectives of the program

Demonstrate the fit between the aims and objectives of the program and the expertise of the consortium members. Also describe the complementarity of the members.

As described above the **ALIVE** team consists of a diverse group of researchers from a wide variety of departments within the three institutions. All team members are experts on topics central to this Flagship; together these unique complementary knowledge areas and skills will allow them to create world-class research and substantial societal impact. In short, **ALIVE** combines expertise in:

- All relevant aspects of **lifestyle**, including physical activity, sleep and diet, and specifically the field of health we will focus on in the first 5 years: brain and mental health.
- Using various types of **data**, such as population-based cohorts (e.g., Generation R Study, Rotterdam Study, GOUD), registries and patient records (e.g., IPCI, Rijnmond Gezond and CBS), clinical studies (e.g., Erasmus Stroke Study, Co-Flow study) and qualitative studies (e.g., Contextmapping, Focusgroup and Participatory Research).
- A range of **statistical tools and infrastructures** for big data analysis, storage options, and combining data sources. Our data scientists, biostatisticians, medical informatics and epidemiologists create a data science network that facilitates medical outcomes research. They will analyze different types of data: quantitative data, such as neuroimaging data, using machine learning and deep learning algorithms, and qualitative data using thematic analyses and human-centered study design.
- Development of **policy and guidelines**. Members of our team are for example part of working groups of the Dutch Health Council for the development of dietary guidelines, advisors on guidelines to prevent eating disorders, and have created platforms for patient groups, e.g., for individuals with intellectual disabilities or patients with arthrosis.

Please note that by working with experts from different fields and institutions, we can combine knowledge on data science with practical skills around legal issues relating to linkage. This is currently often fragmented which can cause risks with regards to data safety and privacy, but also prevents optimal use of resource. Existing infrastructures such as the Research Suite (Erasmus MC), national Health RI initiative (medical data), the ODISSEI project (social science) will therefore be leveraged in this project, to optimize the use of these systems in a meaningful way for all researchers at TU Delft, Erasmus MC and Erasmus University.

4.c Involvement of partners

Indicate which partners are involved, how and in what stage they are involved, as well as how well team members are connected to the external partners in the ecosystem.

A broad network of external partners will be involved in different stages of the project. First and foremost we will create a sounding board of external partners to ensure the outcomes of our

project create true impact. These partners will be invited for yearly Flagship meetings to meet with the consortium, and they will be contacted on a regular basis. Mainly, these partners will be involved in phases of research development, making sure we design our research in a way our findings will be relevant for practice and society. Also, these partners will be of major importance in phases of tool and guidelines development, making sure these tools and guidelines align with their users' needs and expectations. This sounding board will include professionals working in clinics, industry, government, students, patients and research participants. Longstanding partners that will be of high relevance for the **ALIVE** Flagship are for example the Municipality of Rotterdam (Wilma Jansen, knowledge broker Youth), health insurance companies (Menzis, Achmea), industry partners in digital medicine (Big Health Ltd), student societies ('Stichting Student en Leefstijl') and patient/participant representatives ('ReumaNederland', Rotterdam Study participant group). In the future, these and other contacts will also be leveraged to attract additional external funding for the **ALIVE** Flagship.

4.d Governance, organization and collaboration strategy of the consortium

Please describe how the consortium is organized to facilitate collaboration and make optimal use of the expertise of all participants.

To reach our aim to contribute to more effective, individual-based lifestyle interventions across the lifecourse, we need to ensure true convergence within the **ALIVE** team, making optimal use of each person's skills, knowledge and resources. Therefore, the Flagship program will be led by representatives the three institutions, the WPs will be highly linked to facilitate collaboration and involve at least two of the three institutions as key personnel. Moreover, all new positions created with this funding (PhD candidates and post-doctoral researchers), will be joint appointments between at least two of the three institutes, and involve multiple departments. Additionally, we will specifically dedicate one of the vacancies for an overall post-doctoral researcher to ensure all research is executed in an integrated manner and create time for project management duties which will include regular meetings between **ALIVE** members, both within and across WPs. Additionally, we will organize yearly **ALIVE** meetings that are accessible for all researchers of TU Delft, Erasmus MC and Erasmus University, to ensure that convergence is not only reached within the **ALIVE** team but also far beyond.

Annexes (do not count toward maximum number of pages)

I Budget table (including the external funding table)

A pluriannual plan will be required if your proposal is accepted

II Letters of intent and letters of commitment

On headed paper of the entity. Letters of commitment are preferable to letters of intent and letters of intent are preferable to the absence of these letters.

III Letter of support from department head of the main lead

On headed paper of the department or institution.

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Budget formulier: Convergentie Flagship Call

Flagship:	A Lifecourse and Individual-based View on lifestyle to Enhance health
Flagship acronym:	ALIVE
Naam Penvoerder:	Annemarie Luik
Startdatum en looptijd van het project:	01-09-2022 tm 31-08-2027 (60 maanden)
Totaal project budget:	€ 2.000.000,00

R&D Overview: kostensoorten per type onderzoek (hele project periode)

1.	Loonkosten:									2022		2023		2024		2025		2026		2027		Totaal		Totaal per partner
	Partner die de kosten maakt	Functie	Aantal fte	Startdatum	Einddatum	Inschaling	Trede	Maandbedrag	Kosten	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	
	Onderzoeksorganisaties																							
	EMC	OIO	2,00	01-09-2022	01-09-2024	UMC - OIO	0	€ 3.166	€ 237.472	€ 32.304,00		€ 102.796		€ 102.372,00		€ -		€ -		€ -		€ 237.472	€ 0	€ 237.472,00
	EMC	OIO	2,00	01-09-2022	01-09-2026	UMC - OIO	0	€ 3.166	€ 465.445	€ 32.304,00		€ 102.796		€ 117.642,00		€ 125.228		€ 87.474,00		€ -		€ 465.444	€ 0	€ 465.444,00
	EMC	OIO	2,00	01-03-2023	01-03-2027	UMC - OIO	0	€ 3.197	€ 469.908			€ 81.816		€ 113.180,00		€ 122.390		€ 130.646,00		€ 21.876,00		€ 469.908	€ 0	€ 469.908,00
	EMC	Postdoc/Data mana	2,00	01-09-2022	30-08-2026	UMC - 11	0	€ 4.639	€ 668.023	€ 51.308,00		€ 156.674		€ 165.788,00		€ 173.952		€ 120.302,00		€ -		€ 668.024	€ 0	€ 668.024,00
	TUD	OIO	2,00	01-09-2022	01-09-2026	UMC - OIO	0	€ 3.166	€ 465.445	€ 32.304,00		€ 102.796		€ 117.642,00		€ 125.228		€ 87.474,00		€ -		€ 465.444	€ 0	€ 465.444,00
	TUD	OIO	2,00	01-03-2023	01-03-2027	UMC - OIO	0	€ 3.197	€ 469.908			€ 81.816		€ 113.180,00		€ 122.390		€ 130.646,00		€ 21.876,00		€ 469.908	€ 0	€ 469.908,00
	EUR	OIO	2,00	01-09-2023	01-09-2027	UMC - OIO	0	€ 3.228	€ 474.512			€ 32.934		€ 105.074,00		€ 119.626		€ 127.686,00		€ 89.192,00		€ 474.512	€ 0	€ 474.512,00
	EUR	postdoc	2,00	01-03-2023	01-09-2026	UMC - 11	0	€ 4.525	€ 583.748			€ 129.956		€ 163.070,00		€ 171.300		€ 119.422,00		€ -		€ 583.748	€ 0	€ 583.748,00
	Ondernemingen																							
	Onderneming 1							€ -	€ 0													€ 0	€ 0	€ 0,00
	Onderneming 2							€ -	€ 0													€ 0	€ 0	€ 0,00
	Onderneming 3							€ -	€ 0													€ 0	€ 0	€ 0,00
	Onderneming 4																							€ 0,00
	Overige (private) partners																							
	Naam 1							€ -	€ 0													€ 0	€ 0	€ 0,00
	Naam 2							€ -	€ 0													€ 0	€ 0	€ 0,00
	Naam 3							€ -	€ 0													€ 0	€ 0	€ 0,00
	Naam 4							€ -	€ 0													€ 0	€ 0	€ 0,00
	Totaal									€ 3.834.459												Totaal loonkosten		€ 3.834.460

2.	Kosten van materialen en hulpmiddelen:									2022		2023		2024		2025		2026		2027		Totaal		Totaal per partner
	Partner die de kosten maakt	Materiaal / hulpmiddel						Kosten	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind		
	Onderzoeksorganisaties																							
	Organisatie 1																					€ 0	€ 0	€ 0,00
	Organisatie 2																					€ 0	€ 0	€ 0,00
	Organisatie 3																					€ 0	€ 0	€ 0,00
	Organisatie 4																					€ 0	€ 0	€ 0,00
	Ondernemingen																							
	Onderneming 1																					€ 0	€ 0	€ 0,00
	Onderneming 2																					€ 0	€ 0	€ 0,00
	Onderneming 3																					€ 0	€ 0	€ 0,00
	Onderneming 4																					€ 0	€ 0	€ 0,00
	Overige (private) partners																							
	Naam 1																					€ 0	€ 0	€ 0,00
	Naam 2																					€ 0	€ 0	€ 0,00
	Naam 3																					€ 0	€ 0	€ 0,00
	Naam 4																					€ 0	€ 0	€ 0,00
	Totaal									€ 0												Totaal kosten materialen en hulpmiddelen		€ 0

3.	Kosten van gebruik van machines en apparatuur									2022		2023		2024		2025		2026		2027		Totaal		Totaal per partner
	Partner die de kosten maakt	Machine / apparatuur				Prijs per gebruikseenheid		Gebruiks eenheden	Kosten	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	
	Onderzoeksorganisaties																							
	EMC																					€ 0	€ 0	€ 0,00
	Organisatie 2																					€ 0	€ 0	€ 0,00
	Organisatie 3																					€ 0	€ 0	€ 0,00
	Organisatie 4																					€ 0	€ 0	€ 0,00
	Ondernemingen																							
	Onderneming 1																					€ 0	€ 0	€ 0,00
	Onderneming 2																					€ 0	€ 0	€ 0,00
	Onderneming 3																					€ 0	€ 0	€ 0,00
	Onderneming 4																					€ 0	€ 0	€ 0,00
	Overige (private) partners																							
	Naam 1																					€ 0	€ 0	€ 0,00
	Naam 2																					€ 0	€ 0	€ 0,00
	Naam 3																					€ 0	€ 0	€ 0,00
	Naam 4																					€ 0	€ 0	€ 0,00
	Totaal									€ 0												Totaal kosten machines en apparatuur		€ 0

4.					2022		2023		2024		2025		2026		2027		Totaal		Totaal per partner			
					in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind						
	Aan derden verschuldigde kosten		Naam derde, omschrijving kosten			Kosten																
	Partner die de kosten maakt																					
	Onderzoeksorganisaties																					
	Organisatie 1																		€ 0		€ 0	€ 0,00
	Organisatie 2																		€ 0		€ 0	€ 0,00
	Organisatie 3																		€ 0		€ 0	€ 0,00
	Organisatie 4																		€ 0		€ 0	€ 0,00
	Ondernemingen																					
	Onderneming 1																		€ 0		€ 0	€ 0,00
	Onderneming 2																		€ 0		€ 0	€ 0,00
	Onderneming 3																		€ 0		€ 0	€ 0,00
	Onderneming 4																					
	Overige (private) partners																		€ 0		€ 0	€ 0,00
	Naam 1																		€ 0		€ 0	€ 0,00
	Naam 2																		€ 0		€ 0	€ 0,00
	Naam 3																		€ 0		€ 0	€ 0,00
	Naam 4																		€ 0		€ 0	€ 0,00
			Totaal			€ 0													Totaal kosten derden		€ 0	

5.					2022		2023		2024		2025		2026		2027		Totaal		Totaal per partner		
					in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind					
	Publicatie, reis- en verblijfkosten		Naam derde, omschrijving kosten			Kosten															
	Partner die de kosten maakt																				
	Onderzoeksorganisaties																				
	EMC/TUD/EUR		Travel WP Leads and hired personel			€ 45.000			€ 15.000		€ 15.000,00		€ 15.000				€ 45.000		€ 0	€ 45.000,00	
	EMC/TUD/EUR		Service/maintenance research infrastructure			€ 45.000			€ 15.000		€ 15.000,00		€ 15.000				€ 45.000		€ 0	€ 45.000,00	
	EMC/TUD/EUR		Publication costs/Open access			€ 10.000									€ 10.000,00		€ 10.000		€ 0	€ 10.000,00	
	EMC/TUD/EUR		Flagship conferences meetings			€ 30.000							€ 15.000		€ 15.000,00		€ 30.000			€ 30.000,00	
	EMC/TUD/EUR		Data support fees cohort studies			€ 70.000			€ 40.000		€ 30.000,00						€ 70.000		€ 0	€ 70.000,00	
	EMC/TUD/EUR																				
	Ondernemingen																€ 0		€ 0	€ 0,00	
	Onderneming 1																€ 0		€ 0	€ 0,00	
	Onderneming 2																€ 0		€ 0	€ 0,00	
	Onderneming 3																€ 0		€ 0	€ 0,00	
	Onderneming 4																€ 0		€ 0	€ 0,00	
	Overige (private) partners																€ 0		€ 0	€ 0,00	
	Naam 1																€ 0		€ 0	€ 0,00	
	Naam 2																€ 0		€ 0	€ 0,00	
	Naam 3																€ 0		€ 0	€ 0,00	
Naam 4																€ 0		€ 0	€ 0,00		
		Totaal			€ 200.000											Totaal kosten derden		€ 200.000			

6.					€ 4.034.459																€ 4.034.460	
	Totale kosten																					

FINANCIERING - Bijdragen per partner per jaar

in Euro		2022		2023		2024		2025		2026		2027		Totaal		Totaal per partner	
		in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind		
Onderzoeksorganisaties																	
1	EMC	€ 115.916		€ 444.082		€ 498.982		€ 421.570		€ 338.422		€ 21.876		€ 1.840.848	€ 0	€ 1.840.848	
2	TUD	€ 32.304		€ 184.612		€ 230.822		€ 247.618		€ 218.120		€ 21.876		€ 935.352	€ 0	€ 935.352	
3	EUR			€ 162.890		€ 268.144		€ 290.926		€ 247.108		€ 89.192		€ 1.058.260	€ 0	€ 1.058.260	
4	EMC/TUD/EUR			€ 70.000		€ 60.000		€ 45.000		€ 25.000				€ 200.000	€ 0	€ 200.000	
Totale bijdrage onderzoeksorganisaties		€ 148.220	€ 0	€ 861.584	€ 0	€ 1.057.948	€ 0	€ 1.005.114	€ 0	€ 828.650	€ 0	€ 132.944	€ 0	€ 4.034.460	€ 0	€ 4.034.460	
Ondernemingen																	
4	Onderneming 1													€ 0	€ 0	€ 0	
5	Onderneming 2													€ 0	€ 0	€ 0	
6	Onderneming 3													€ 0	€ 0	€ 0	
...	Onderneming 4													€ 0	€ 0	€ 0	
Totale bijdrage ondernemingen		€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	
Overige (private) partners																	
7	Naam 1													€ 0	€ 0	€ 0	
8	Naam 2													€ 0	€ 0	€ 0	
9	Naam 3													€ 0	€ 0	€ 0	
...	Naam 4													€ 0	€ 0	€ 0	
Totale overige bijdrage		€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	
									</								

GRANTED FUNDS					
Funder	Awardee	Title	Period	In-cash/kind	EUR
NIH	Annemarie Luik	Multidimensional Sleep Health and mental health	2021-2025	In-cash	150000
CSC	Trudy Voortman	Physical activity and sedentary behavior in elderly	2020-2024	In-kind	65000
EU Joint Programme Neurodegenerative Diseases	Trudy Voortman	Lifestyle and brain health	2020-2023	In-cash	50000
Erasmus Trustfund	Bram Wouterse	Living with dementia	2022-2024	In-cash	150000
Alzheimers Association	Daniel Bos	Trajectories of Vascular Disease in Dementia	2022-2024	In-cash	100000
Innovative Medicines Initiative	Peter Rijnbeek	EHDEN (maintaining and developing a Common Data Model for health data)	2018-2023	In-cash	100000
NWO	Manon Kok	Widely available sensors help us find our way indoors	2022-2024	In-cash	250000
ZonMW	Hans Bussmann	Develop and evaluate a new and interactive therapy aid based on accelerometers	2020-2023	In-cash	150000
Care organizations of the HA-ID consortium	Alyt Oppewal	Healthy Ageing and Intellectual Disabilities: 10-year follow-up study	2018-2023	In-cash	426000
ZonMW	Alyt Oppewal	In beweging (implementatie bij mensen met een verstandelijke beperking)	2021-2024	In-cash	67000
ZonMW	Marienke van Middelkoop Dieuwke Schiphof	The (cost)effectiveness of a combined lifestyle intervention in overweight or obese patients with knee osteoarthritis in primary care	2021-2024	In-cash	625000
TOTAL					2133000
SUBMITTED/IN PREPARATION					
Funder	Awardee	Title	Period	In-cash/kind	EUR
CSC	Annemarie Luik	Maternal sleep and child and mother health	2022-2026	In-kind	65000
SNF	Annemarie Luik Trudy Voortman	Lifestyle and Muculoskeletal health	2022-2026	In-cash	200000
Parkinson Fond	Kamran Ikram Annemarie Luik Trudy Voortman	Social and lifestyle factors in relation to Parkinson	2022-2026	In-cash	250000
TOTAL					515000
FUTURE FUNDS					
During the project, new project proposals will be developed for future funding from funding bodies such as ZonMW, ERC and NIH, but also charities and industry partners.					

To: Convergence Health & Technology 2022

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Our reference Convergence Flagship
Date March 11, 2022

Dear sir/madam,

With this letter I, as the department head of the Department of Epidemiology at Erasmus MC University Medical Center Rotterdam, confirm my full support for the Convergence Flagship Program application '**A Lifecourse and Individual-based View on lifestyle to Enhance health (ALIVE)**' led by Dr. Annemarie Luik, Dr. Trudy Voortman, Dr. Jos Kraal and Dr. Bram Wouterse.

The **ALIVE** Flagship Program approaches the important topic of lifestyle and health from the Convergence Health & Technology mission, a topic at the intersection of health and technology that needs to be addressed urgently. Moreover, it will lead to convergence across a large number of departments across the three universities by involving a diverse group of highly motivated and talented researchers, particular at early and mid-career levels to create sustainability long after the first 5-years of this initial project. Their expertise and experience in the field of health and technology is outstanding, demonstrated by their ability to extract external funding, excellent research and societal impact. Additionally, research infrastructure and new methods benefiting all researchers of the three institutes will be developed within this five-year period to achieve convergence far beyond the **ALIVE** Flagship Program. Access to data, appropriate facilities, technological and logistical support will be made available to newly hired researchers for this Flagship Program.

Overall, I have high expectations for this Flagship Program and their long-term contribution to the convergence. Please contact me if you require any further information.

With kind regards,



M. Arfan Ikram, MD, PhD
Professor and Chair Epidemiology

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Dimitris Rizopoulos, PhD
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Bruno Ch. Stricker, MD, PhD
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Sonja A. Swanson, ScD
Causal Inference

Meike W. Vernooij, MD, PhD
Population Imaging

Trudy Voortman, PhD
Nutrition & Lifestyle Epidemiology